

Nonlinear Dynamics & Chaos HW1

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Technion

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1 Quadratic Map

Consider the one-parameter family of maps

$$x_{n+1} = x_n^2 + c, \quad c \in \mathbb{R} \quad (1)$$

1. Draw cobweb plots of the map for $c = 0.4$ and $c = 0$. Briefly describe the qualitative behavior you read from each cobweb
2. Find all fixed points as a function of c and classify their linear stability
3. Find the values of c at which bifurcations occur and write the type of bifurcation for each value
4. You should have found a value of c at which a fixed point loses stability but no new fixed point is born. Show that instead a period-2 orbit emerges. find p and q explicitly as functions of c .
5. Find the range of c for which the 2-cycle is stable.
6. Sketch a partial bifurcation diagram in the c, x plane for $c \in [-1.5, 0.5]$. Mark the fixed point branches and the 2-cycle branch, and mark the bifurcation values from (c).

1.1 cobwebs

The function we'll use to draw is $f(x) = x^2 + c$. On the left cobweb, we can see that the behavior is unbound

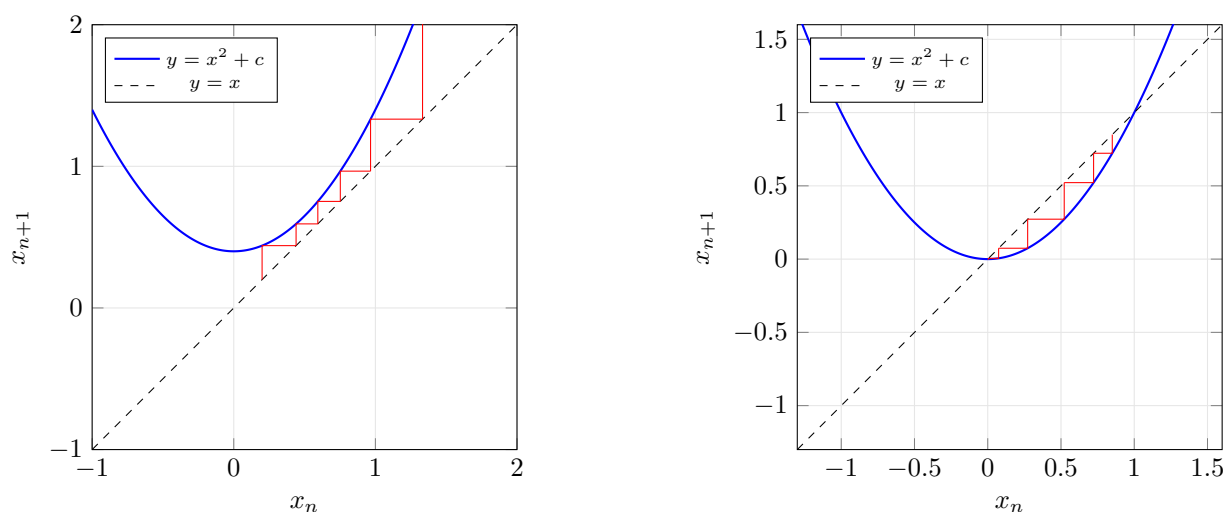


Figure 1 – Cobweb diagrams ($c = 0.4$ left, $c = 0$ right).

and will tend to infinity. On the right cobweb, $x = 1$ is a source and $x = 0$ is a stable sink. points beyond $x = 1$ will go to infinity. points before $x = 0$ will converge to $x = 0$.

1.2 Fixed points and Stability

To find the fixed points we solve

$$x^2 - x + c = 0 \quad (2)$$

whose solutions are

$$x_{\pm} = \frac{1 \pm \sqrt{1-4c}}{2} \quad (3)$$

looking at the derivative at those points and restricting to only real values

$$f'(x_{\pm}) = 1 \pm \sqrt{1-4c} = \begin{cases} f'(x_+) = 1 + \sqrt{1-4c} \geq 1 \\ f'(x_-) = 1 - \sqrt{1-4c} \end{cases} \quad (4)$$

to check the second point's stability we require that

$$-1 < 1 - \sqrt{1-4c} < 1 \quad (5)$$

which is satisfied for

$$-\frac{3}{4} < c < \frac{1}{4} \quad (6)$$

in summary, x_+ is always unstable, and x_- is stable for $-\frac{3}{4} < c < \frac{1}{4}$.

1.3 Bifurcations and Types

Using the last section we see two points:

1. $c = \frac{1}{4}$. Both fixed points combine and disappear for $c > \frac{1}{4}$. When that happens the derivative is exactly +1 so the graph of f is tangent to the diagonal, meaning this is a tangent bifurcation.
2. $c = -\frac{3}{4}$. No new fixed point is created so this is a period-doubling bifurcation.

1.4 period-2 orbit

period-2 points are where $f^2(x) = x$ so in our case

$$0 = (x^2 + c)^2 + c - x = x^4 + 2cx^2 - x + c^2 + c = 0 \quad (7)$$

but since we have two fixed points from the original $f(x)$ we can treat them as solutions and factor them out using polynomial division giving

$$f^2(x) - x = (x^2 - x + c)(x^2 + x + c + 1) \quad (8)$$

which means the 2-cycle is the roots of the second element

$$p, q = \frac{-1 \pm \sqrt{-4c-3}}{2} \quad (9)$$

we can notice that for $c = -\frac{3}{4}$ both roots coincide meaning the 2-orbit emerges from the fixed point associated with that value of c .

1.5 Stability range

Taking the derivative

$$(f^2)'(p) = f'(f(p)) \cdot f'(p) = f'(q) \cdot f'(p) = 4pq \quad (10)$$

then using Vieta on the second element we have

$$pq = c + 1 \quad (11)$$

so the multiplier is

$$\mu = 4c + 4 \quad (12)$$

and to have stability

$$-1 < 4c + 4 < 1 \implies -\frac{5}{4} < c < -\frac{3}{4} \quad (13)$$

1.6 bifurcation diagram

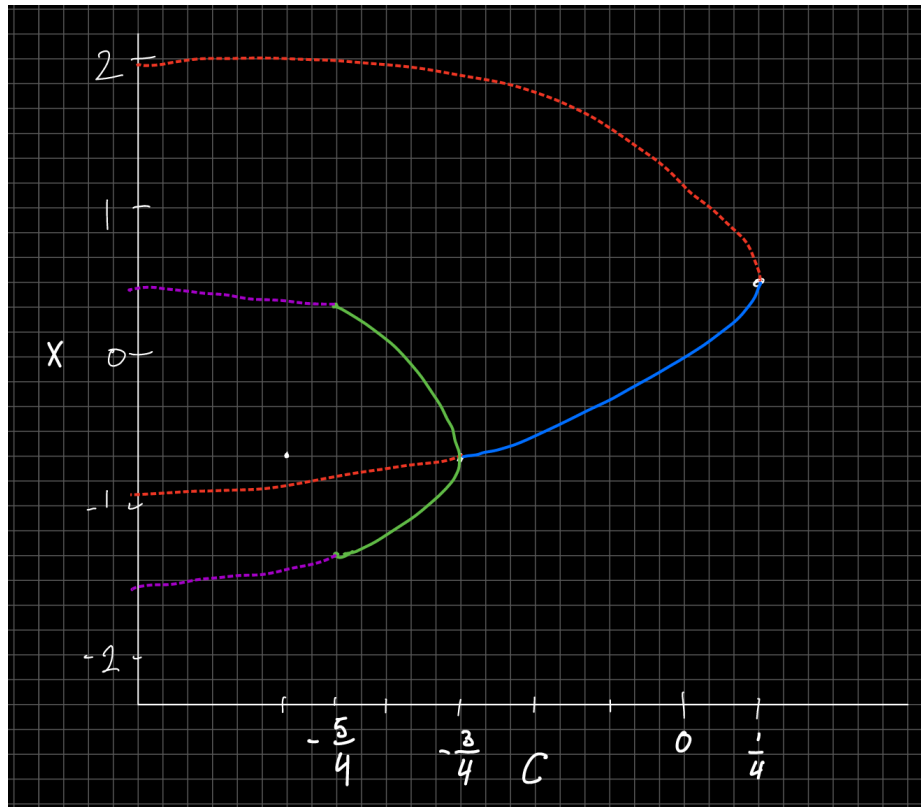


Figure 2 – bifurcation diagram. Solid lines are stable, dashed are unstable. The tangent bifurcation is seen at $c = 0.25$, and the flip at $c = -0.75$. At $c = -1.25$ the 2-cycle destabilises.

2 Newton Method as an iterated map

1. Write down the Newton map explicitly for $g(x) = x^2 - 4$. Show that it has exactly two fixed points that coincide with the roots of g , and that both are superstable in the sense of the definition above.
2. Starting from $x_0 = 1$ iterate and report the first few iterates $x_1, x_2, \dots$ until the result agrees with $x^* = 2$ to your tool's precision
3. Return to a general twice differentiable g and let x^* be a root, $g(x^*) = 0$. Show that x^* is a superstable fixed point of the newton map if $g'(x^*) \neq 0$. What happens if $g'(x^*) = 0$?